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Deterministic quantum light emitters in DNA origami–engineered molecule–MoS₂ hybrids

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Abstract

The functionalization of atomically-thin transition metal dichalcogenides (TMDs) with organic molecules is a promising approach for realizing nanoscale optoelectronic devices with tailored functionalities, such as quantum light generation or *p-n* junctions. However, achieving precise control over the molecules' positioning on the 2D material remains a significant challenge. Here, we overcome the limitations of solution- and vapor-deposition methods and use a DNA origami placement technique to spatially arrange thiol molecules on a chip surface at the single-molecule level with high assembly yields. We successfully integrated MoS₂ monolayers with micron-scale thiol–origami patterns, creating quantum-emitting sites from thiol-induced localized excitons in MoS₂. Our work lays a foundation for the chemical control of quantum emitters in atomically-thin semiconductors and enables the design and production of ultracompact 2D devices for quantum technologies.

Introduction

Solid-state single-photon emitters (SPEs) provide a scalable and robust pathway for achieving quantum memory in quantum networking applications^{1,2}. Among these, SPEs based on atomically-thin TMDs^{3,4} offer several advantages: their emission wavelengths fall within the visible/near-infrared range, their two-dimensional nature facilitates integration into nanoscale devices, and they exhibit high brightness and narrow linewidths⁵. SPEs in TMDs can be created through defects induced by strain^{6–10} or point defects^{10,11}. These defects are typically fabricated via local bending^{6–8,12}, mechanical stretching^{9,13}, or ion- or electron beam irradiation^{10,11,14}. However, a common limitation of these techniques is the difficulty in deterministically creating SPEs at specific locations with reproducible properties. Therefore, developing scalable, deterministic, and precise methods for SPE fabrication remains a significant challenge.

Molecular surface functionalization presents an alternative approach to engineering quantum emitters in low-dimensional materials. For instance, functionalization with organic molecules has been used to introduce quantum defects¹⁵ and achieve single-photon emission¹⁶ in carbon nanotubes. For atomically-thin materials such as graphene and TMDs, functionalization with organic molecules has also inspired the design of novel device architectures, including high-performance photo-detectors^{17,18}, *p-n* junctions^{19–21}, single-molecule sensors^{22,23}, and catalytic systems²⁴. In these devices, organic molecules deposited on the surface of 2D material introduce charge doping²⁵ or energy-transfer states²⁶, leading to changes in their optical²⁷, electronic, thermoelectric²⁸, and magnetic properties²⁹. However, assembling such hybrid devices at the molecular scale involves several challenging requirements, including nanometer-precise control of molecule positioning, and versatility in the choice of molecule. Current fabrication methods offer limited control over the homogeneity and positioning of organic molecules, as well as the stability of the final hybrid structure, which hinders progress in this field. Chemical surface functionalization via the non-covalent³⁰ or covalent³¹ bonding of adsorbed molecules are commonly used to fabricate hybrid structures. However, the

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solution processing methods such as spin-coating, drop-casting, or vapor-based methods lack full control over molecule concentration and position on a 2D material surface. Conventional photolithography processes including photochemical reactions, chemical etching or laser writing^{32–34} can only pattern layers of organic molecules on top of 2D material.

Structural DNA nanotechnology, particularly DNA origami self-assembly^{35,36}, enables bottom-up fabrication of highly defined, complex two- and three-dimensional nanostructures with single-nanometer resolution^{37–42}. DNA origami is widely used as a molecular scaffold, enabling sub-nanometer precision in positioning molecules and nanoparticles^{43–48}. Kershner et al. developed a DNA origami placement (DOP) technique by combining DNA origami self-assembly with lithographic nanopatterning, allowing for site- and shape-selective deposition of individual DNA origami objects into arrays on patterned substrates⁴⁹. DOP overcomes the limitations of top-down lithography and gives access to high-yield placement and arrangement of individual nanoscale components such as metallic nanoparticles^{50–52} and organic dyes^{53,54} into precise arrays and patterns. Importantly, DOP yields of more than 90%^{52,54} surpass the single-molecule binding efficiency of 37% imposed by Poisson statistics on traditional single-molecule deposition methods^{55,56}. Progressing from these achievements, an approach that allows the fabrication of molecular patterns for chemical control of quantum emitters in TMDs is highly desirable.

Here, we demonstrate programmable 2D material–organic molecule hybrids with high efficiency and spatial precision. By dry-stamp transferring micron-scale, chemical-vapor-deposited MoS₂ monolayers onto the chips patterned with DNA origami triangles bearing thiol molecules, we rationally design hybrid systems that form arrays of single-photon-emitter ensembles with nanosecond lifetimes and high spectral and intensity stability. We achieve an ~90% yield in quantum-emitter placement with a mean positioning accuracy of ~13 nm. Our method provides a platform for precisely engineering the electronic properties of 2D materials at the nanoscale and opens a path toward producing miniaturized hybrid inorganic–organic devices with enhanced performance.

Results

Fabrication design

Thiol-terminated molecules are widely used for functionalizing MoS₂ both in aqueous solutions and in air^{57–59}. Building on this approach, we employ thiol groups in our study and precisely position them on lithographically patterned substrates using DNA origami triangles with 127-nm-long outer edges as “molecular adaptors” (Fig. 1a). An atomically-thin, single-crystal

MoS₂ flake is subsequently transferred on top of the air-dried thiol-origami pattern, resulting in thiol molecules binding to the MoS₂ (Fig. 1b).

We used a variant of the “Rothemund triangle” as it has been successfully positioned on lithographically patterned substrates before⁶⁰. The DNA triangle carries 18 3'-adenine “anchor” strands, each 20 nt long, extending from the triangle along its inner edges for functionalization with 19 nt thymine “linker” strands bearing on their 5' ends a single thiol group (the design of the anchor strands is presented in Supplementary Note 1 and Fig. S1).

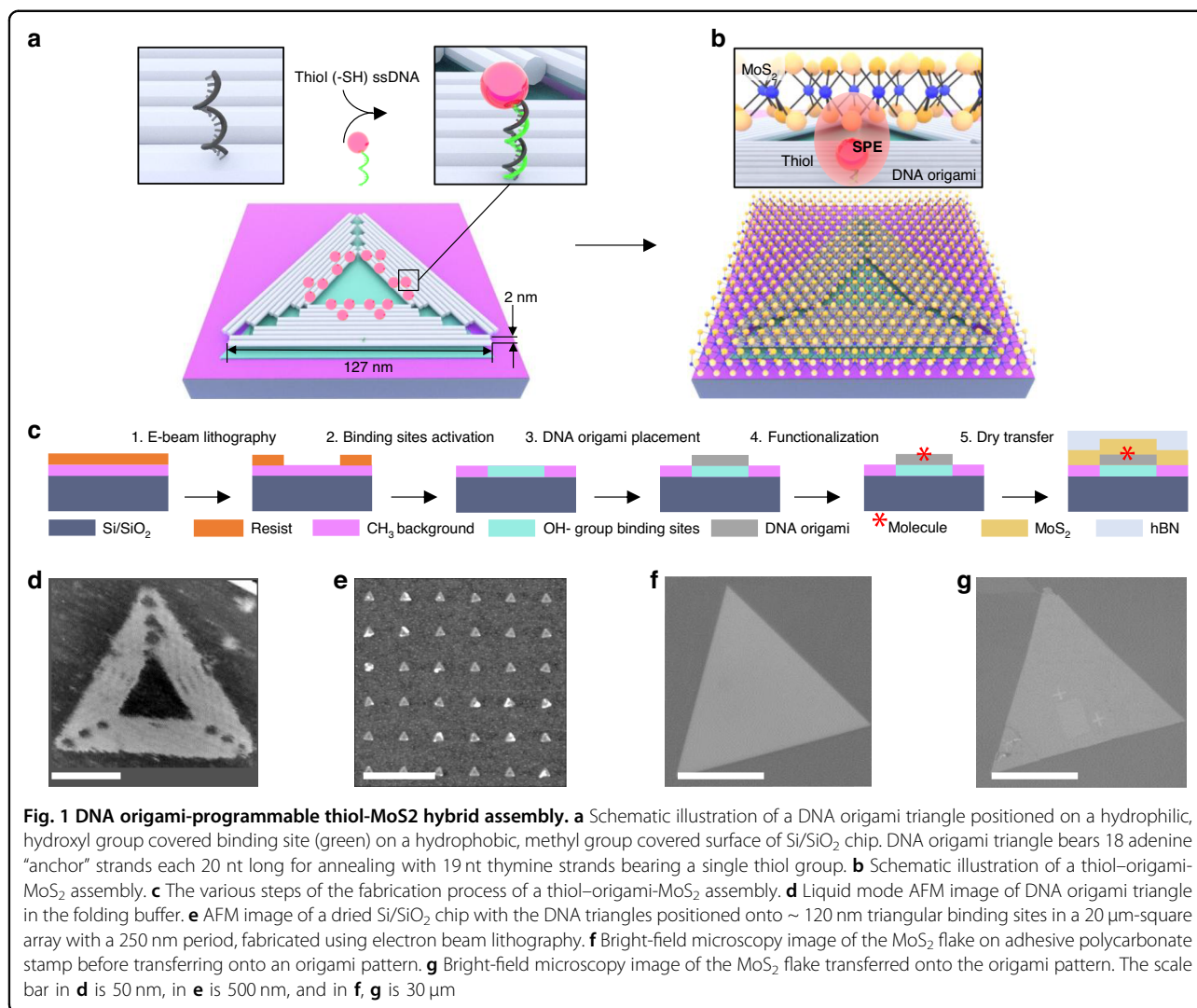
The various steps of the fabrication process are illustrated in Fig. 1c. First, DNA triangles, designed *in silico*^{61,62} and folded in buffer containing MgCl₂ (Fig. 1d), were deposited onto specific binding sites that were created *via* electron-beam (e-beam) patterning of negatively charged hydroxyl groups (green) within a background of hydrophobic methyl groups (pink) on a Si/SiO₂ substrate. Negatively charged origami binds strongly to the charged spots via positively charged Mg²⁺ ions from buffer solution while the hydrophobic, thus passivated area, binds origami poorly.

Substrate preparation and DNA triangle placement were performed according to previously established protocols^{52,60}. In brief, Si/SiO₂ chips with square arrays of triangular binding sites were incubated with thiol-functionalized DNA triangles at 25 °C for 1 h in a Tris buffer with 35 mM MgCl₂. After placement, the Si/SiO₂ chips with the thiol-origami patterns were air-dried. Atomic force microscopy (AFM) revealed a typical height of 2 nm for the dried DNA triangles, consistent with the height of a DNA duplex (Fig. 1e). In this study, we used square arrays of triangular binding sites with various periods ranging from 170 nm to 1000 nm. Detailed information on the thiol-origami pattern designs, along with AFM characterization and height profiles of the air-dried patterns, is provided in Supplementary Note 2 and Supplementary Figs. S2 and S3. Consistent with our prior findings⁵², the yield of single origami binding exceeded 90% for all patterns, independent of the array period.

Finally, triangular single-crystal monolayers of MoS₂ covered by hexagonal boron nitride (hBN) flakes (Fig. 1f) were transferred onto a thiol-origami pattern using an adhesive polycarbonate stamp, followed by annealing under vacuum. Figure 1g depicts a representative optical microscope image of the resulting hybrid assemblies, showing the origami pattern fully covered with MoS₂.

Optical properties of MoS₂ on thiol-origami pattern

We first aimed to confirm binding between MoS₂ and the origami-bound thiols. To that end, we fabricated dense, periodic square arrays of thiol-functionalized DNA triangles with a 170 nm period, resulting in ~40 nm spacing between adjacent DNA triangles (see Fig. 2 and insets



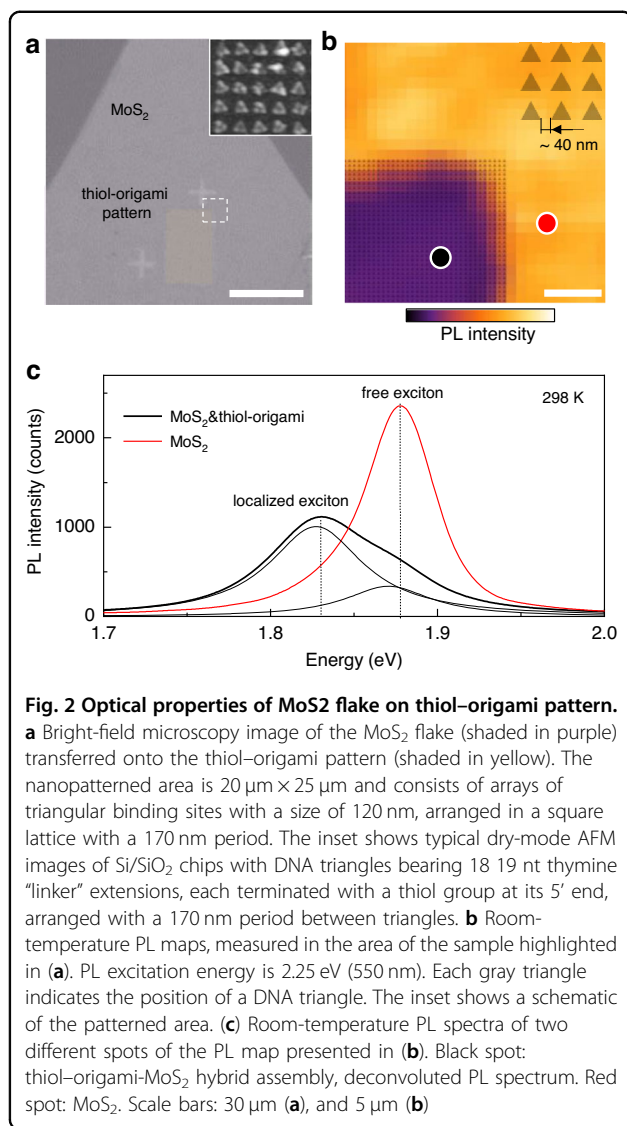
in Fig. 2a, b). The thiol-origami pattern design and its AFM characterization are presented in Supplementary Fig. S2. We then transferred a micron-scale, atomically thin MoS₂ flake onto this pattern (Fig. 2a). A control sample without thiol functionalization was also fabricated to rule out interactions between DNA origami and MoS₂.

Figure 2b depicts room-temperature (RT) photoluminescence (PL) maps of MoS₂ transferred onto the thiol-origami pattern. Both the non-patterned MoS₂ and the MoS₂ deposited on non-thiolated origami patterns exhibited similar PL spectra, characteristic of MoS₂ monolayers on dielectric substrates (Fig. 2c, Supplementary Fig. S4). Specifically, the PL spectra contain a single peak at 1.88 eV with a full-width at half-maximum (FWHM) of 49 meV, corresponding to the lowest excitonic resonance, known as the A exciton⁶³. This exciton corresponds to the momentum-direct transition at the K-valleys in the Brillouin zone of the MoS₂ monolayer. Thiol-origami patterning of MoS₂ introduced a new PL

peak at 1.83 eV (Fig. 2b, c). This new peak is red-shifted by 50 meV compared to the A exciton transition and exhibits half the intensity and a 29% broader FWHM of 68 meV. Notably, this red-shifted peak cannot be attributed to trion states, as the monolayer MoS₂ trion binding energy of 34 meV^{64,65} is less than the observed 50 meV energy splitting between the A exciton and the red-shifted peak. This splitting is more consistent with localized exciton states, such as those created by helium ion irradiation¹¹. We hypothesize that thiol binding to MoS₂ leads to the formation of localized states, potentially serving as the origin of single-photon emission^{66–70}.

Single photon emission

To optically resolve PL signal variations directly corresponding to the positions of individual thiol-origami structures, we patterned MoS₂ with a square array of DNA triangles with a 1000 nm period (Fig. 3a) and recorded a PL map. Figure 3b depicts the ratio of PL



intensities for the localized and free exciton peaks at 1.83 eV and 1.88 eV, respectively. The emission intensity of the free exciton is significantly reduced at all lattice sites, indicating efficient exciton capture or funneling into localized emitters. This confirms that the thiolated strands on each individual DNA triangle bind to MoS₂, forming trapping sites for excitons and resulting in emission at an energy 50 meV lower than that of the free exciton in MoS₂.

Next, we investigated the emission properties of thiol-functionalized MoS₂ at 4 K. Figure 3c depicts a typical PL spectrum from an individual site, marked by the red circle in Fig. 3b. At low temperature, both free and localized MoS₂ emission peaks are blue-shifted by ~50 meV^{71,72}. The broad PL of localized excitons resolves into several sharp peaks. Figure 3d presents the second-order photon-correlation function $g^{(2)}(\tau)$ for a single sharp peak (marked

in gray in Fig. 3c). The measured second-order correlation function exhibits a prominent dip at zero-time delay with $g^{(2)}(0) = 0.31$. The value of $g^{(2)}(0)$, well below the 0.5 threshold, demonstrates single-photon emission from our thiol-origami patterned MoS₂. The reduced $g^{(2)}(0)$ contrast (~0.3) mainly arises from incomplete spectral filtering of nearby emission peaks, as well as a residual background contribution from the free A-exciton, both of which lead to additional uncorrelated photons and thereby limit the measured antibunching.

Time-resolved PL measurements reveal that the PL lifetimes of the SPE are typically on the order of nanoseconds (Fig. 3e). This experimental observation is in stark contrast to helium-ion-induced emitters in MoS₂, which exhibit microsecond lifetimes¹¹, and aligns more closely with the SPEs in WSe₂, which have nanosecond lifetimes^{66–68}. Remarkably, the quantum emitters demonstrate minimal photobleaching, blinking, and spectral diffusion, indicating robust performance (Fig. 3f and Supplementary Fig. S5). We estimate the quantum yield of the quantum emitters, η , to be ~10% using saturation measurements under pulsed excitation (see Supplementary Note 3, Supplementary Fig. S6, and Supplementary Table 1).

Furthermore, we measured low-temperature PL spectra for 33 thiol-origami binding triangles. We observed localized emission at all 33 positions, corresponding to the thiol-origami locations. Representative PL spectra from three of these positions are shown in Supplementary Fig. S7. Each spectrum exhibits four to six emission lines that are absent in non-patterned areas. This demonstrates that patterning with thiol-origami enables the deterministic induction of quantum defects in atomically thin MoS₂. All measured defect emitters exhibited low-temperature localized emission red-shifted by ~50 meV from the free exciton peak, consistent with the RT emission. Out of the 33 measured thiol-origami sites, 29 exhibited single-photon emission peaks, resulting in a SPE placement yield of 0.88 with a 95% confidence interval of 72.9–95.2% (see Supplementary Fig. S7a,b for PL data and statistical analysis).

Note that the power-dependent PL of all single-photon peaks exhibits near-linear scaling, which is inconsistent with the presence of multi-exciton complexes. In addition, the absence of antibunching in cross-correlation measurements between peaks within the same thiolated triangle suggests that these emissions originate from distinct quantum emitters, likely associated with different localization sites (see Supplementary Figs. S8 and S9, respectively). We further extract linewidth for the PL peaks showing single-photon emission across multiple triangles (Fig. 3c, Supplementary Fig. S7c–e and Supplementary Table 2). A mean FWHM of 0.79 ± 0.19 meV confirms that the coherence-relevant linewidths consistently

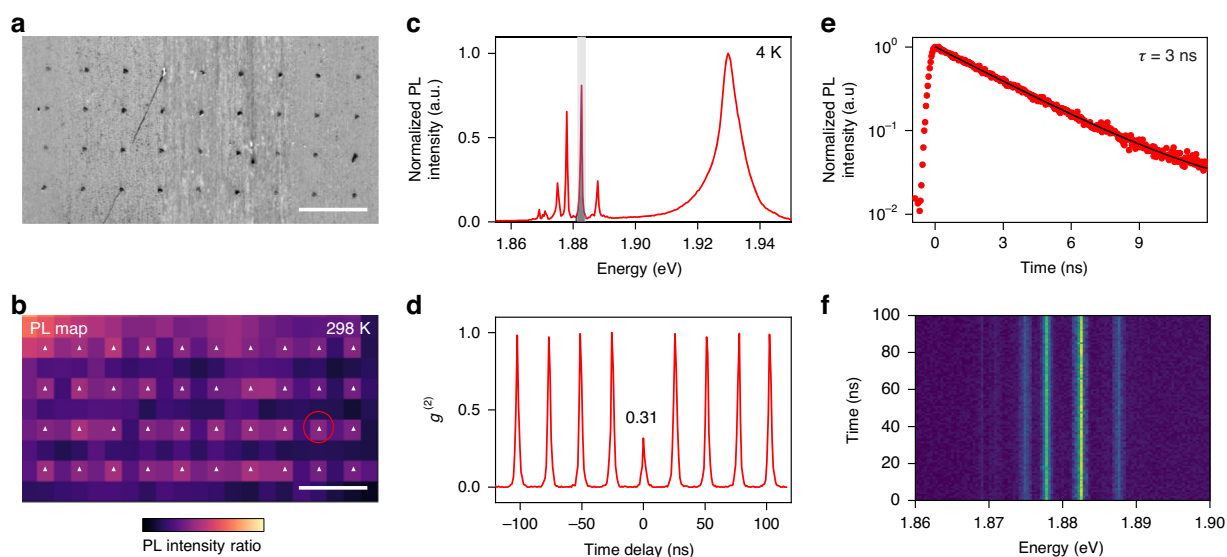


Fig. 3 DNA origami-programmable single-photon emission in MoS₂-thiol molecule hybrids. **a** Typical dry-mode AFM images of Si/SiO₂ chips with DNA triangles bearing 18 19-nt thymine “linker” extensions, each terminated with a thiol group at its 5′ end, arranged with a 1000 nm period between triangles. **b** Map of the ratio of localized to free exciton PL peaks in a thiol-origami-MoS₂ hybrid assembly. Each white triangle indicates the position of a DNA triangle. **c** Low-temperature (4 K) PL spectrum from the spot indicated by the red circle in **(b)**. The PL peak at 1.883 eV is fitted with a Lorentzian function (dark blue) and has a FWHM of 0.57 meV. **d** Correlation function of the localized defect state marked by the gray bar in **(c)**, showing second-order coherence at zero-time delay with $g^{(2)}(0) = 0.31$. **e** Time-resolved PL of localized defect state marked by the gray bar in **(c)** showing a single-exponential decay with a decay time of 3.099 ± 0.001 ns. **f** Low-temperature PL intensity of localized states as a function of measurement time. All measurements were performed under non-resonant excitation at 2.25 eV. Scale bars: 2 μ m **(a, b)**

remain well below 1 meV across different triangles. To quantify spectral inhomogeneity, we also analyzed the single photon emission peak energies across multiple triangles. The emitters exhibit a mean peak position of 1.880 eV with a standard deviation of 3 meV (see Supplementary Table 2). Although spectral inhomogeneity may limit applications requiring indistinguishable photons, the relatively narrow distribution suggests that further control of defect conditions could reduce this spread.

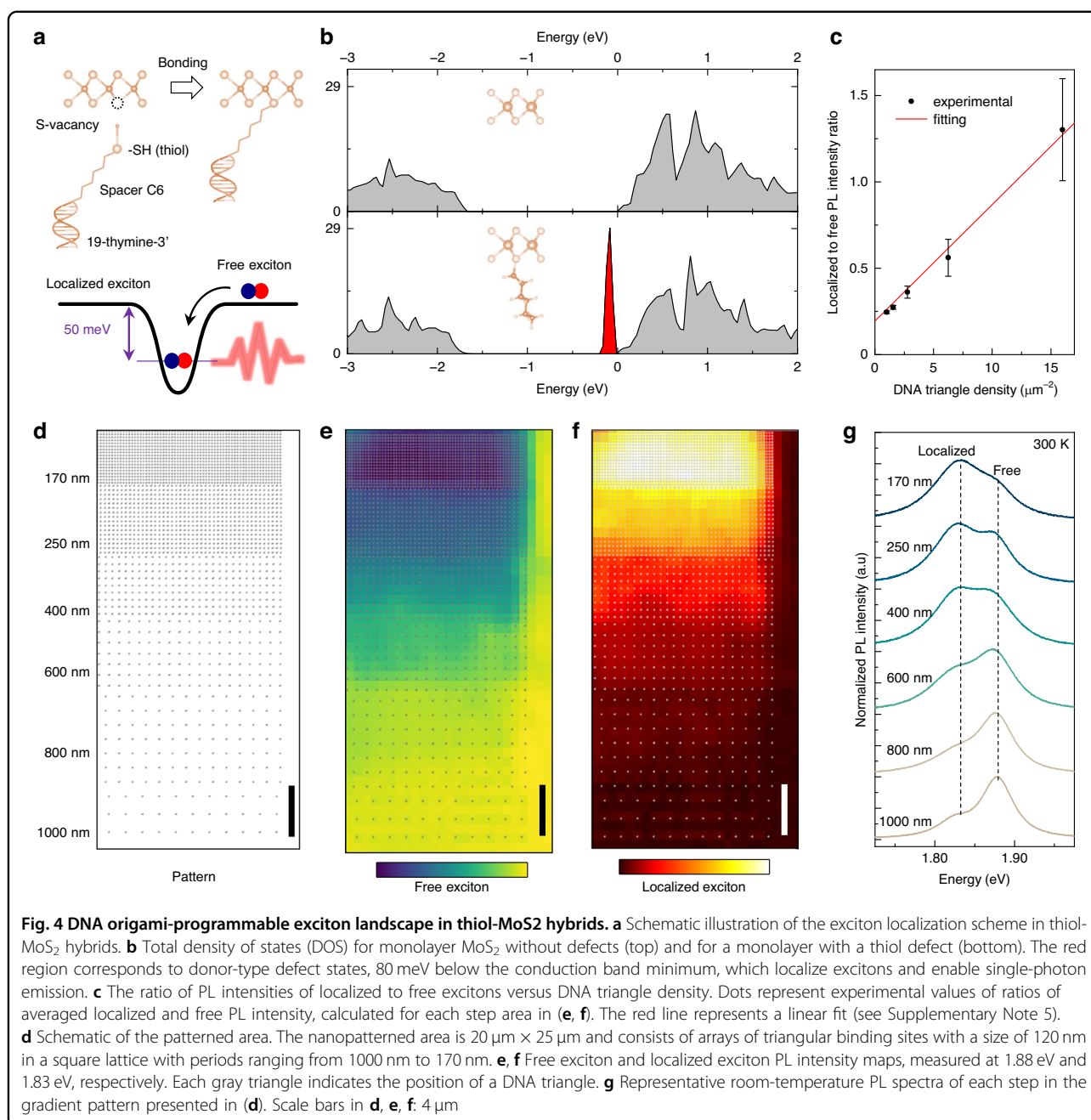
The positioning accuracy of our technique is predominantly determined by the positioning accuracy of individual thiol molecules by DOP technique, which is ~ 13.8 nm (1σ) (calculation of positional accuracy is presented in Supplementary Note 4 and Supplementary Fig. S10).

Origin of defect states

It is commonly assumed that thiol-derivative molecules bind to sulfur vacancies in MoS₂^{59,73–75}. Consistent with this assumption, we attribute the binding to chemisorption of thiols to sulfur vacancies in MoS₂, as illustrated in Fig. 4a. Reported sulfur vacancy densities in MoS₂ vary significantly depending on the experimental conditions and evaluation technique, ranging from $\sim 10^{13}$ cm⁻² as measured by super-resolved optical mapping⁵⁹ to 10^{15} cm⁻² as reported by TEM and STM investigations^{76–78}. However, the relatively high density of these

vacancies compared to our thiolated strands (estimated at $\sim 10^{11}$ cm⁻², with each DNA triangle carrying 18 thiolated strands) still ensures efficient binding. Because we observe pronounced changes in the PL spectra from the thiol-functionalized regions, we conclude that chemisorption at sulfur vacancies is the more realistic mechanism. In contrast, physisorption on the MoS₂ surface would not sufficiently modify the optical response, as energy transfer in this case is inefficient due to the non-resonant nature of the interaction between thiol molecule and exciton in MoS₂.

As shown in Fig. 4a, we assume that the thiol molecule creates a trapping potential for excitons near the binding site, with a depth in the order of 50 meV. We estimate this depth using a microscopic approach, based on the known energy splitting between the free exciton (1.88 eV) and the localized exciton (1.83 eV). To further support our concept, we performed density functional theory (DFT) calculations for both pristine monolayer MoS₂ and monolayer MoS₂ functionalized with a sulfur atom bonded to a -C₆H₁₃ tail molecule (see “Methods” for DFT calculations details). We present the ab initio results in Fig. 4b, the top and bottom panels show the total density of states calculated using a plane-wave basis set in the Vienna Ab-initio Simulation Package (VASP). For all plotted figures, the zero-energy reference is set at the conduction band minimum to enable straightforward



identification and comparison of defect-level positions. The top panel shows the typical behavior of pristine MoS₂, whereas the bottom panel clearly reveals the formation of an in-gap state. Similar donor-like states have been both theoretically predicted and experimentally observed for sulfur vacancies, impurities, and other defect types in TMDs^{11,79–81}. It is well established that such donor defect states can effectively trap excitons, forming donor-bound excitons (DX states)^{79,81,82}. In this scenario, neutral or charged donors create a local potential that confines the exciton, namely D⁰X and D⁺X complexes.

These donor-bound exciton complexes can efficiently emit light, resulting in single-photon emission. Notably, both neutral and charged complexes likely contribute to the multipeak structures observed in the low-temperature PL spectra in Fig. 3c and Supplementary Fig. S7. A second possible explanation is the presence of different molecular binding conformations at sulfur vacancies, specifically the various possible orientations of the -C₆H₁₃ spacer molecule relative to the monolayer.

Our PL lifetime measurements of the 3 ns also suggest that we are dealing with shallow donors. In our case,

shallow donor states trap the exciton in a relatively small potential well, which likely preserves a comparatively short lifetime. In contrast, for helium-ion-induced emitters arising from deep donor states, microsecond lifetimes have been observed as reported in ref. ¹¹. In such systems, the microscopic nature of the photon emission can differ significantly from that of free-exciton emission.

Scalability of the functionalization approach

To assess the scalability and density limits of our approach, we fabricated a matrix of 4750 individual thiol–origami arranged in a square array with periods decreasing gradually from 1000 to 170 nm (Fig. 4c–e; see Supplementary Fig. S3 for AFM characterization). Due to the diffraction-limited optical resolution of $\sim 1\ \mu\text{m}$, we could not spatially resolve PL signals from individual lattice sites when the thiol–origami periods ranged from 170 to 800 nm; instead, we observed relatively homogeneous PL signals in these regions. Near-field optical techniques⁸³ can be further used to resolve PL signal from individual lattice sites in these patterns. PL maps of free and localized excitons are shown in Fig. 4e and 4f, respectively. The highest-density pattern with 170 nm period exhibits near-complete quenching of free exciton emission and strong localized exciton emission (Fig. 4g).

We calculated the average intensity of localized and free exciton emission for each step of the pattern. Figure 4c shows the ratio of localized to free exciton emission as a function of DNA triangle density. Our experimental values are well fitted by a linear relationship (see Supplementary Note 5). Based on this observed linear dependence, we propose that the SPE placement yield could be independent of the pattern period and may reach 88%, even for the densest patterns. Then, using our densest patterns, up to 5×10^5 single-photon light sources could be easily achieved on a single $1000\ \mu\text{m}^2$ MoS₂ equilateral triangle with a side length of $50\ \mu\text{m}$, demonstrating the remarkably high yield and density of our quantum light sources.

Discussion

Although our present demonstration relies on electron-beam lithography, which is not suitable for parallel wafer-scale manufacturing, established surface patterning approaches such as nanoimprint lithography^{60,84,85} and colloidal lithography^{52,86} have achieved electron-beam-lithography-quality single-origami placement while enabling parallel, low-cost, large-area fabrication. These methods therefore offer practical pathways to extend our technique from the current proof of concept and toward wafer-scale fabrication.

Overall, our DNA origami-programmed functionalization of MoS₂ is site-selective, non-destructive, and uses inexpensive materials that are easy and safe to handle. By

employing our functionalization approach, we were able to introduce quantum emitters in MoS₂ with molecular-level precision, previously achievable only with destructive top-down techniques such as focused-ion beam irradiation. These results highlight DNA origami as a unique and versatile tool for customizing MoS₂ materials with precisely targeted localized quantum defects acting as SPEs. Further tuning of the number of thiol molecules per origami and exploring different organic molecules provides a route to improve single-photon purity⁸⁷ and generate chiral quantum light⁸⁸.

In summary, we tuned the optical properties of monolayer MoS₂ via functionalization with thiol molecules, precisely positioned on chip surfaces using a DOP technique. By transferring MoS₂ monolayer onto micron-scale thiol–origami patterns, we realized arrays of quantum-emitting sites, each hosting an ensemble of SPEs. Complementary DFT calculations indicated that the observed single-photon emission originates from exciton localization induced by the thiol–MoS₂ chemical bonding. Moreover, by adjusting the square-lattice period of thiol–origami patterns, we achieved unprecedented control over the density of these localized excitons. Looking forward, our technique can be used for precise functionalization of a wide range of 2D materials, including TMDs, graphene, and other 2D van der Waals structures. In nanophotonics, our hybrid 2D organic-inorganic structures will enable exciton landscape engineering, where DNA origami can precisely control the number and density of localized defects for next-generation circuits and quantum emitters¹¹ for quantum communications. In addition to nanophotonic devices, our hybrid approach may be useful for nanoelectronic applications⁸⁹ and any heterogeneous fabrication process requiring the integration of molecules or nanoparticles with 2D materials at high spatial accuracy and orientation control.

Materials and methods

DNA origami design, preparation and purification

The “sameside sharp triangle” design is adapted from ref. ⁶⁰. We modified 18 staples from the original design by extending them with 20-nucleotide adenine extensions on the 3' end. The positions of these extended staples are shown in Supplementary Fig. S1. These 20-nucleotide adenine-extended staples serve as linkers that bind to a 19-nucleotide thymine strand bearing a single thiol molecule on its 5' end in a “shear” configuration.

DNA origami triangles were folded by mixing scaffold strands (7249 nucleotides long M13mp18 single-stranded DNA) with an excess of staple strands in folding buffer (10 mM Tris, 1 mM EDTA, 12.5 mM MgCl₂, pH 8.35). The scaffold DNA was mixed with all staples except for 18 staples, which were replaced with modified versions containing 20-nucleotide adenine “anchor” strands at the

3' ends and thiolated 19-nucleotide thymine strands. The final concentrations in the folding buffer were: 20 nM scaffold strand, 100 nM staple strands, 500 nM polyA-modified staples (IDT, 200 μ M) and 2000 nM thiolated polyT strands (Biomers, HPLC purified, 100 μ M). The samples were annealed in a PCR machine (Biometra TRIO Thermal Cycler, Analytik Jena) and purified from excess staples by Amicon filtration as described in ref. ⁵². After Amicon purification, the concentration of triangles typically ranged from 70 to 100 nM, with a recovery yield of 40–50%. Scaffold strands were produced from M13 phage replication in *Escherichia coli*. All chemicals were obtained from Sigma Aldrich unless otherwise stated.

Preparation of the substrates and DNA origami placement

Si/SiO₂ substrates were patterned using electron-beam lithography, following protocols from ref. ⁵² with slight modifications. A 4-inch Si/SiO₂ wafer with a 100 nm thermal oxide layer (Microchemicals) was diced into 1 cm $\times$ 1 cm chips. Clean chips were primed with 10 mL of hexamethyldisilazane (HMDS) in a 4 L desiccator. The priming time was optimized to maintain a Si/SiO₂ surface contact angle of 70°–75° after HMDS deposition. Triangular binding sites, each 120 nm per side, where individual DNA triangles could bind, were patterned into poly(-methyl methacrylate) resist using electron-beam lithography. The chips were then developed with a 1:3 solution of methyl isobutyl ketone and isopropanol (IPA). The HMDS in the developed areas was removed using O₂ plasma for 6 s in a plasma cleaner (PICO). The resist was stripped by ultrasonication in N-methyl pyrrolidone at 50 °C for 30 min. The substrates were briefly rinsed with 2-propanol, dried in a nitrogen stream, and used immediately.

DNA triangles were bound to the patterned substrates as previously described in refs. ^{52,60}. A 20–60 μ L drop of freshly folded and purified DNA origami was deposited onto the surface of the chips in placement buffer (5 mM Tris, 35 mM MgCl₂, pH 8.35). After incubation for 1 h at RT in a 100% humidity incubator, excess DNA origami was removed from the surface by performing eight buffer replacement steps and purifying with 0.1% Tween 20. After this step, chips with absorbed DNA origami were air-dried using an ethanol dilution series as previously described⁶⁰.

MoS₂ monolayer synthesis and transfer

Monolayers of MoS₂ were synthesized by the vapor phase chalcogenization method⁹⁰ on thermally oxidized SiO₂/Si substrates with oxide thickness of 285 nm. A three-zone furnace system (Carbolite Gero) equipped with a 1-inch quartz tube was used for growth under ambient pressure. In order to precisely place MoS₂ monolayers onto molecule-origami patterns, a dry-

transfer method with PDMS/PC stamps was used⁹¹. An hBN flake with a thickness of $\sim$ 100 nm was firstly picked up at 50 °C, followed by a triangular monolayer MoS₂ at 145 °C. The entire stack was subsequently released from the stamp onto a SiO₂/Si target substrate with molecule-origami pattern at a temperature of 180 °C. The sample was then soaked in chloroform solution for 5 min to remove polycarbonate residues, cleaned by acetone and isopropanol and annealed under ultrahigh vacuum for 12 h.

Characterization techniques and data analysis

UV-vis absorption measurements were performed with a NanoDrop ND-1000 Spectrophotometer (Thermo Scientific). Tapping-mode AFM of dried Si/SiO₂ substrates with triangular DNA origami was carried out on a Dimension ICON AFM (Bruker). OTESPA silicon tips (300 kHz, Veeco Probes) were used for imaging in air. Images are analyzed with the Software Gwyddion.

Hyperspectral PL imaging was performed using a custom-built scanning confocal microscope. The sample was mounted on piezo-stepping units (ANPxyz101, attocube system) for precise positioning with respect to the confocal spot of an apochromatic objective (LT-APO/532-RAMAN, attocube system). Continuous-wave diode lasers with a power of 2 μ W were used for excitation to prevent sample damage. Spectrally sharp short-pass and long-pass filters (Semrock) were employed in the excitation and detection paths, respectively, to eliminate laser light. The sample's PL signal was dispersed by a monochromator (Acton SpectraPro 300i, Roper Scientific) with a 300 grooves/mm grating and detected using a Peltier-cooled charge-coupled device (CCD, Andor iDus 416). We subtracted the hBN background emission from the raw PL data.

The low-temperature PL measurements were performed in a close-cycle cryostat (attoDRY800, attocube system) with a base temperature of 4 K, using the same scanning confocal microscope as for the RT measurements. The second-order photon-correlation $g^{(2)}(\tau)$ measurements were carried out using a standard Hanbury–Brown and Twiss setup. The sample was excited by a supercontinuum laser (SuperK EXTREME, NKT Photonics) tuned to 550 nm, and the emission was detected with a pair of silicon avalanche photodiodes (τ -SPAD, PicoQuant). Detection events were recorded and correlated with a time-correlated single-photon counting (TCSPC) module (PicoHarp300, PicoQuant).

DFT calculations

The electronic structure calculations of low-dimensional crystalline lattices were performed using VASP^{92–94} within DFT^{95,96} and Periodic Boundary Conditions (PBC). To treat the exchange-correlation GGA-

PBE functional⁹⁷ was used. Plane-wave basis set coupled with projector augmented wave (PAW) method with the cutoff energy of 700 eV were used to describe electronic distribution. The Brillouin zone was sampled by $12 \times 12 \times 1$ k-point mesh for the monolayer, and $6 \times 6 \times 1$ k-point mesh for 3×3 supercell case. All calculations were performed in PBC with 14 Å vacuum intervals to prevent any interfacial interactions. During the optimization procedure, the maximum force acting on atoms less than 0.005 eV/Å was used as a stopping criterion for structural minimization. For all plotted figures, we construct the density of states using a triangular broadening with a width of 70 meV.

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Author contributions

I.M., A.B., and S.Z. designed this study. I.M. designed, assembled, and purified DNA origami samples, performed DNA origami placement experiments, and conducted AFM measurements and data analysis with assistance from E.E. and C.S. Iu.M. and A.B. performed DFT calculations. Z.L. performed MoS₂ monolayer synthesis and, together with I.M., transferred it onto the origami patterns. K.W. and T.T. performed hBN synthesis. S.Z. and Z.L. performed optical spectroscopy. I.M., A.B., and S.Z. analyzed the data and wrote the manuscript with input from A.H. and T.L. Z.L. and S.Z. contributed equally to this work.

Data availability

The authors confirm that the data supporting the findings of this study are available within the article and its supplementary materials.

Conflict of interest

The authors declare no competing interests.

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